Econ 182: Sections

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Unemployment Fluctuations

$$\theta = \frac{V}{U} = \frac{V}{H-L}$$

$L \uparrow \Rightarrow \theta \uparrow$

positive correlation

Question 1

In the United States, which correlation do we observe over the business cycle?

(A) Unemployment level and labor market tightness are positively correlated.

(B) Employment level and labor market tightness are positively correlated.

(C) Unemployment level and vacancies are positively correlated.

(D) Unemployment level and employment level are positively correlated.

(E) Unemployment level and labor force participation are positively correlated.

(F) None of the above.

Question 2

θ & L positive correlation

In the matching model with fixed wage, which type of shocks can generate the correlation described in the previous question?

Labor Supply : Shock to H

Labor demand : Shock to a

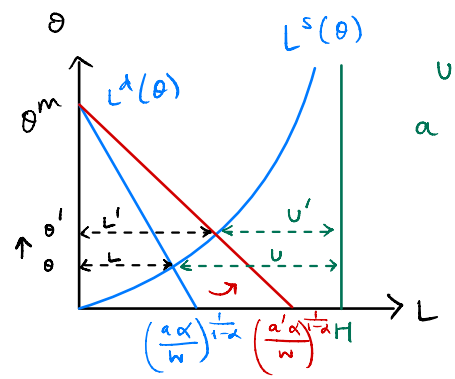
(A) Shocks to labor productivity

(B) Shocks to the size of the labor force

(C) Shocks to the disutility from unemployment

(D) Shocks to monetary policy

(E) No shocks can generate such correlation



$U' < U$, $L' > U$

$a \uparrow \Rightarrow L \uparrow, \theta \uparrow$

$U \downarrow$

$a' > a$

Question 3

Consider a matching model with surplus sharing and a linear production function. Assume that the value of unemployment is $z > 0$ and that the bargaining power of firms is 1. Then an increase in labor productivity a leads to:

$\beta = 0$

(A) Higher tightness and lower unemployment

(B) Lower tightness and higher unemployment

(C) Higher tightness and higher unemployment

(D) Lower tightness and lower unemployment

(E) No effect on tightness and unemployment

From $w = (1-\beta) z + \beta a (1 + r\theta)$

$\Rightarrow w = z$

Labor demand

$$L^d(\theta) = \left[\frac{\alpha a}{w(1+\tau(\theta))^{\alpha}}\right]^{\frac{1}{1-\alpha}}$$

Raise both sides to $1-\alpha$,

$$(L^d)^{1-\alpha} = \frac{a}{z(1+\tau(\theta))}$$

Now $\alpha = 1$ because of linear prod. fn.

$$\underbrace{(L^d)^0}_{1} = \frac{a}{z(1+\tau(\theta))}$$

$$1 = \frac{a}{z(1+\tau(\theta))}$$

If $a \uparrow$, $\theta \uparrow$ since $\tau(\theta)$ is increasing in θ.

$a \uparrow \Rightarrow \theta \uparrow$ and $L \uparrow$, $U \downarrow$

$\theta = \frac{V}{U}$

$$\epsilon_x^a = \frac{d \ln a(x)}{d \ln x} \quad \text{[elasticity definition]}$$

Question 4

Let $c(x) = a(x) \times b(x)/d(x)$. Let ϵ_x^a, ϵ_x^b, ϵ_x^c, and ϵ_x^d be the elasticities of the functions a, b, c, and d with respect to x. Then:

(A) $\epsilon_x^c = \epsilon_x^a \times \frac{\epsilon_x^b}{\epsilon_x^d}$

(B) $\epsilon_x^c = \frac{a(x)}{d(x)}\epsilon_x^a + \frac{b(x)}{d(x)}\epsilon_x^b$

(C) $\epsilon_x^c = \ln(a(x)) + \ln(b(x)) - \ln(d(x))$

(D) $\epsilon_x^c = \epsilon_x^a + \epsilon_x^b - \epsilon_x^d$

(E) None of the above

$$c(x) = \frac{a(x) \times b(x)}{d(x)}$$

1 Take logs on both sides

$$\ln c(x) = \ln a(x) + \ln b(x) - \ln d(x)$$

2 Take derivative wrt $\ln x$,

$$\frac{d \ln c(x)}{d \ln x} = \frac{d \ln a(x)}{d \ln x} + \frac{d \ln b(x)}{d \ln x} - \frac{d \ln d(x)}{d \ln x}$$

$$\boxed{\varepsilon_x^c = \varepsilon_x^a + \varepsilon_x^b - \varepsilon_x^d}$$

Question 5

Let $f(x, y)$ be a function of x and y. Let $\partial f/\partial x$ and $\partial f/\partial y$ be the partial derivatives of the function f with respect to x and y. Let $\epsilon_x^f = \partial \ln(f)/\partial \ln(x)$ and $\epsilon_y^f = \partial \ln(f)/\partial \ln(y)$ be the partial elasticities of the function f with respect to x and y. Then the infinitesimal change in f generated by infinitesimal changes in x and y satisfies:

(A) $df = \epsilon_x^f \cdot dx + \epsilon_y^f \cdot dy$

(B) $d \ln f = \epsilon_x^f \cdot dx + \epsilon_y^f \cdot dy$

(C) $d \ln f = [\partial f/\partial x]dx + [\partial f/\partial y]dy$

(D) $df = \epsilon_x^f \cdot d \ln x + \epsilon_y^f \cdot d \ln y$

(E) $d \ln f = \epsilon_x^f \cdot d \ln x + \epsilon_y^f \cdot d \ln y$

(F) None of the above

$$df(x,y) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

TOTAL DERIVATIVE

1 $$d \ln f(x,y) = \underbrace{\frac{\partial \ln f}{\partial \ln x}}_{\varepsilon_x^f} d \ln x + \underbrace{\frac{\partial \ln f}{\partial \ln y}}_{\varepsilon_y^f} d \ln y$$

SHORTCUT

$$\boxed{d \ln f(x,y) = \varepsilon_x^f \, d \ln x + \varepsilon_y^f \, d \ln y}$$

2 Alternate way:

$$d \ln f = \frac{1}{f} df = \frac{1}{f}\left[\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy\right] = \underbrace{\frac{x}{f}\frac{\partial f}{\partial x}}_{\frac{\partial \ln f}{\partial \ln x}} \underbrace{\frac{dx}{x}}_{d \ln x} + \underbrace{\frac{y}{f}\frac{\partial f}{\partial y}}_{\frac{\partial \ln f}{\partial \ln y}} \underbrace{\frac{dy}{y}}_{d \ln y}$$

Question 6

Let $c(x) = [b \cdot a(x)]^d$, where $a(x) > 0$ and $b > 0$ and $d < 0$. Let ϵ_x^c and ϵ_x^a be the elasticities of the functions c and a with respect to x. Then:

(A) $\epsilon_x^c = [b \cdot \epsilon_x^a]^d$

(B) $\epsilon_x^c = d \cdot [\epsilon_x^a + b]$

(C) $\epsilon_x^c = [b + d] \cdot \epsilon_x^a$

(D) $\epsilon_x^c = d \cdot \epsilon_x^a$

(E) $\epsilon_x^c = b \cdot \epsilon_x^a$

(F) $\epsilon_x^c = d \cdot [b \cdot a(x)]^{d-1}$

(G) None of the above

1 Take logs on both sides,

$$\ln c(x) = d \ln b + d \ln a(x)$$

2 Take derivative wrt $\ln x$,

$$\frac{d \ln c(x)}{d \ln x} = d \underbrace{\frac{d \ln b}{d \ln x}}_{0} + d \frac{d \ln a(x)}{d \ln x}$$

b is CONSTANT

$$\boxed{\varepsilon_x^c = d \cdot \varepsilon_x^a}$$

$$d \ln y = \frac{dy}{y}$$

$$\boxed{\frac{d}{dy} \ln y = \frac{1}{y}}$$

Question 7

Let $c(x) = a(x) + b$, where $a(x) > 0$ and $b > 0$. Let ϵ_x^c and ϵ_x^a be the elasticities of the functions c and a with respect to x. Then

(A) $\epsilon_x^c = \epsilon_x^a$

(B) $\epsilon_x^c = \epsilon_x^a + b$

(C) $\epsilon_x^c = \frac{a(x)}{c(x)} \epsilon_x^a$

(D) $\epsilon_x^c = \frac{b}{c(x)} \epsilon_x^a$

(E) $\epsilon_x^c = \frac{a(x)}{b} \epsilon_x^a$

(F) $\epsilon_x^c = \frac{a(x)}{c(x)} \epsilon_x^a + \frac{b}{c(x)}$

(G) None of the above

1 Take logs on both sides,

$$\ln c(x) = \ln (a(x) + b)$$

2 $$d \ln c(x) = d \ln (a(x) + b))$$

$$= \frac{1}{a(x) + b} d[a(x) + b]$$

$$= \frac{1}{c(x)} [d a(x) + \underbrace{db}_{0}]$$

$$= \frac{a(x)}{c(x)} \underbrace{\frac{d a(x)}{a(x)}}_{d \ln a(x)}$$

$$\frac{d \ln c(x)}{d \ln x} = \frac{a(x)}{c(x)} \frac{d \ln a(x)}{d \ln x}$$

$$\boxed{\varepsilon_x^c = \frac{a(x)}{c(x)} \varepsilon_x^a}$$